IMPERIAL COLLEGE LONDON

EE4-16
EE9-AC6

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2020

MSc and EEE PART IV: MEng and ACGI

ANALOGUE SIGNAL PROCESSING

Thursday, 14 May 10:00 am

Time allowed: 3:00 hours

There are SIX questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P. Georgiou
Second Marker(s) : E. Rodriguez-Villegas

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Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n : Weak Inversion Slope factor.

g_m*: Transconductance.*

1.

a) Describe in your own words what the benefits are of using current mode rather than voltage mode.

[5]

b) *"Analogue circuits are better for signal processing than digital circuits"*. Prove this statement from first principle MOSFET noise equations and discuss in your own words for which scenarios this is true.

[10]

c) For a given mixed mode sensing system, analogue signals need to be digitized for data transmission. What is the best strategy to implement such a system which minimizes power consumption? You are required to describe this in your own words and show an example of an optimum system.

[5]

2. This question concerns translinear circuits.

a) State and derive the translinear principle (TLP) for a loop of MOS transistors working in weak inversion and discuss the limitations. Give all assumptions you make.

[5]

b) For the circuit shown Figure 2.1, I_{X1} and I_{X2} are input currents whereas I_Z is the output current. Express the current I_Z as a function of the currents I_{X1} and I_{X2} and the transistor area K. When $I_{X1}=I_{X2}=I$, determine the value of the transistor area K for which $I_Z=I$.

[10]

c) Figure 2.2 shows a translinear analogue multiplier. Derive its transfer function and show how this circuit can be modified to improve its gain. Derive the differential gain for a two stage multiplier consisting of the improved circuit.

[5]

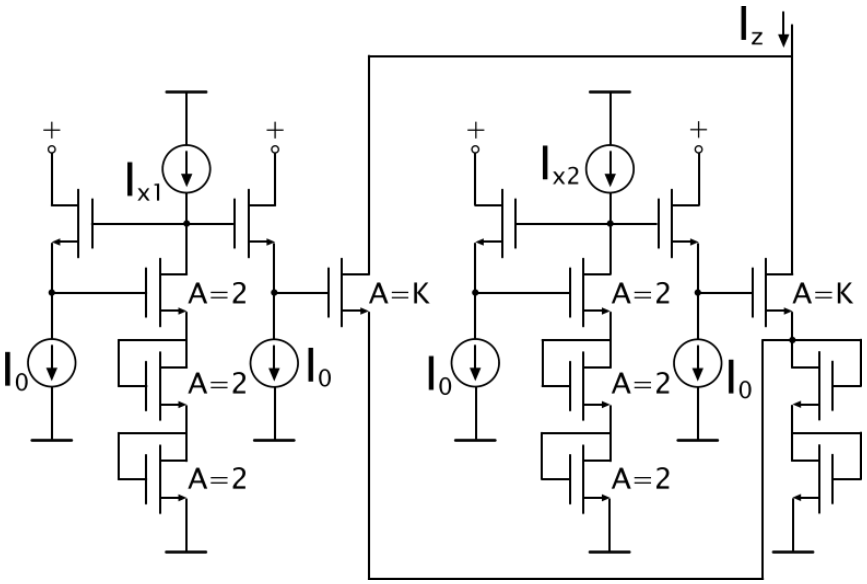


Figure 2.1

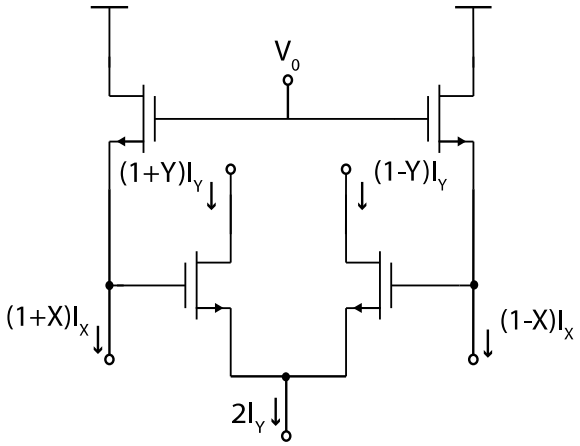


Figure 2.2

3.

a) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_T \tanh(V_c / nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f() composed of just I_{in}, I_1 and I_T.

[6]

b) The transfer function of a second order topology has been decomposed into the following state-space equations:

$$\dot{X_1} = -\left(\frac{\omega_0}{2Q}\right) X_1 - \omega_0 \left(1 - \frac{1}{4Q^2}\right) X_2 + \omega_0 U$$
$$\dot{X_2} = -\left(\frac{\omega_0}{2Q}\right) X_2 + \omega_0 X_1$$
$$Y_1 = X_1$$
$$Y_2 = X_2$$

whereby Y_1 and Y_2 are the outputs, X_1 and X_2 are the state-variables and U is the input.

i. Show that the output Y_1 can be used to implement a second order lowpass transfer function and the output Y_2 can be used to implement a "two-pole one-zero" second order transfer function.

[4]

ii. By using the mappings below, show how these state space equations can be mapped to non-linear log-domain design equations. State any assumptions you make.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

[5]

iii. With these log-domain design equations, sketch a schematic of the final log domain topology which realises the two outputs Y_1 and Y_2 using weak inversion MOS transistors.

[5]

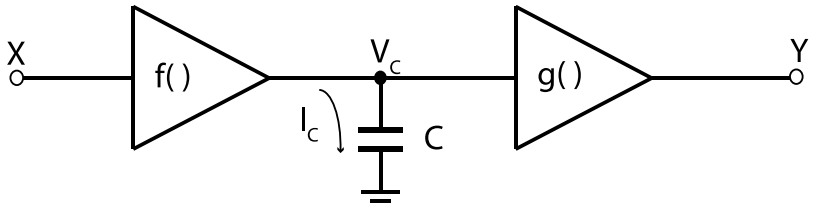


Figure 3.1

4.

a) Figure 4.1 depicts a switched current memory cell. Using this memory cell, design a unit delay cell proving a z^{-1} function and derive the equation for its operation.

[4]

b) Figure 4.2 depicts a FIR filter architecture. Using your unit delay cell from part 4.a sketch a full circuit schematic that implements this function. You may assume the current is unidirectional and use only pMOS current mirrors to copy the current. The coefficients $b_{0\text{-}3}$ must be tunable. Hint: they can be represented by current ratios I_{b0}/I_0, I_{b1}/I_0, I_{b2}/I_0 and I_{b3}/I_0, where I_0 is a MOSFET current in weak inversion.

[8]

c) Figure 4.3 shows a schematic of a switched current integrator and its associated timing diagram. Derive the transfer function of the switched current integrator and find its time constant. You may assume that the switches are ideal and that the transistors sizing is as follows: $\left(\frac{W}{L}\right)_{M_1} = \left(\frac{W}{L}\right)_{M_2}$, $\left(\frac{W}{L}\right)_{M_4} = b_4\left(\frac{W}{L}\right)_{M_1}$ and $\left(\frac{W}{L}\right)_{M_3} = b_3\left(\frac{W}{L}\right)_{M_1}$.

[8]

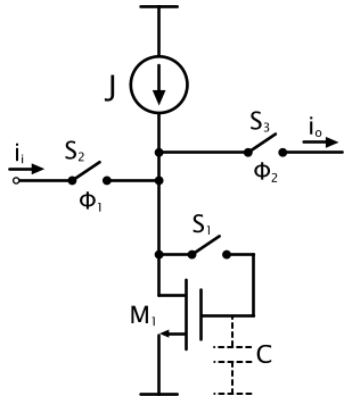


Figure 4.1: Switched current memory cell

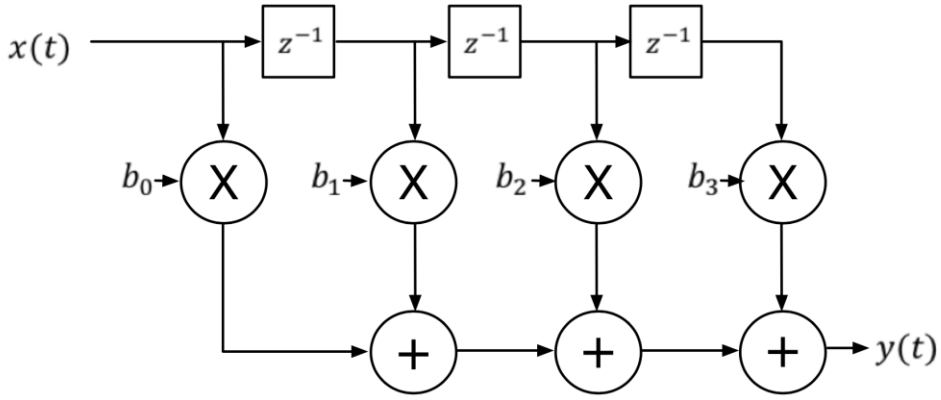


Figure 4.2: FIR filter architecture

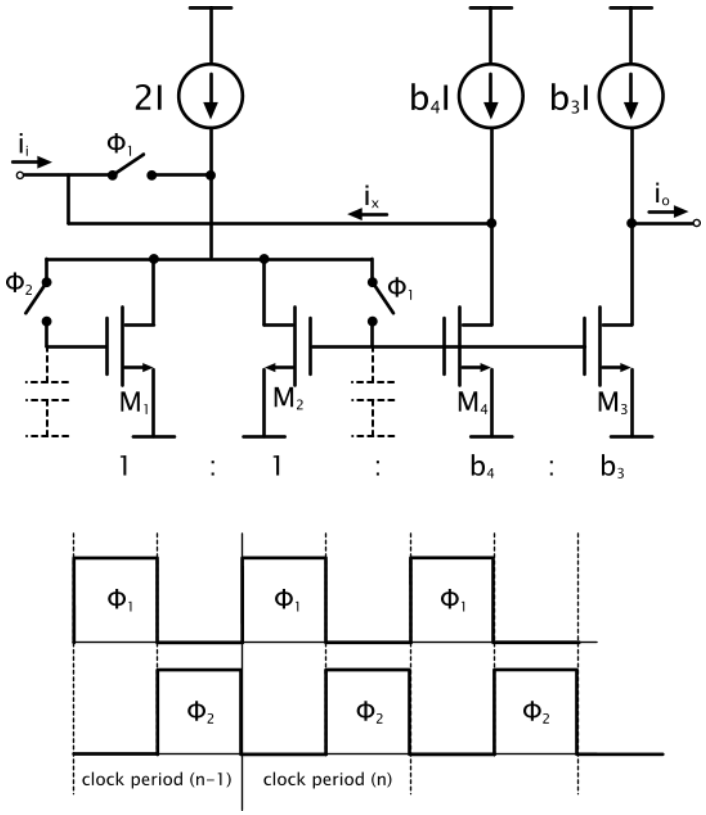


Figure 4.3: Switched current integrator

5. a) Figure 5.1 shows a NMOS sampling switch to be used in a switched capacitor circuit. The parameters of M1 are $(W/L)_1 = 30\mu m/1\mu m$, $C_{ox} = 17$ fF/μm^2, $V_{th} = 0.3$ V, CK = $V_{DD} = 1.8$ V and $C_H = 0.1$ pf. The total charge in the inversion layer, when the switch is on, is given by:

$$Q_{ch} = WLC_{ox}(V_{DD} - V_{in} - V_{th})$$

i) Explain the errors caused due to charge injection and clock feedthrough. Derive equations that show how the output voltage will change due to these errors, assuming the charge injection is split equally between the source and the drain.

[4]

ii) Using the output voltage equations derived in 5i), calculate the total change in output voltage due to charge injection and the total change due to clock feed-through. Based on your result, specify which is more significant. You may assume the overlap capacitance $C_{ov} = 0.08$ fF/m and an input voltage $V_{in} = 1$ V.

[3]

iii) Show that the output voltage contribution due to charge injection results in a non-unity gain error and constant offset. Plot a graph indicating the value of the offset and the slope.

[3]

b) Figure 5.2 shows a circuit diagram of a differential switched capacitor non-inverting amplifier.

i) Explain the necessity of bottom plate sampling in this design and draw a schematic of the amplifier indicating which sides of the capacitors are the bottom plates.

[5]

ii) Derive the transfer function of the Figure 5.2 and also derive an equation to show the change in output voltage during phases Φ_1 and Φ_2.

[5]

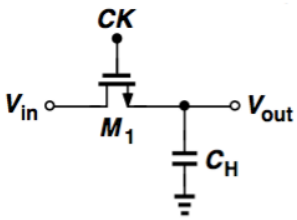


Figure 5.1

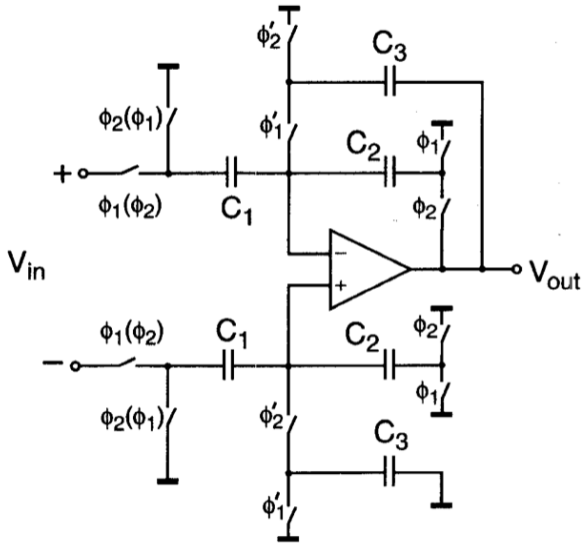


Figure 5.2

6. Figure 6.1 shows one of the most popular topologies in the field of neuromorphic electronics. Stages are denoted by letters A-D, transistors as M1-M13 and bias voltages as Vb1-Vb3.

a) Describe the role of both conductance-based and integrate-and-fire silicon models in your own words. State which category this circuit belongs to.

[3]

b) Sketch the expected output of the circuit. Annotate on the sketch and describe in one sentence the refractory period.

[2]

c) Describe the operation of the positive feedback and state which transistors are involved.

[3]

d) Identify and justify 3 ways to increase the refractory period of the circuit.

[6]

e) One of the main advantages of neuromorphic electronics is the compatibility with Address Event Representation (AER). The diagram of an AER system is represented in Figure 6.2 for an array of neuron pixels. The system operation relies on the use of two signals: a request from the sender neuron and an acknowledge from the AER logic to allow a next signal.
i) Indicate and justify how to complement the neuron topology of Figure 6.1 to generate a digital request signal by adding two transistors or more.

ii) Indicate and justify how to complement the neuron topology of Figure 6.1 to allow a digital acknowledge input by adding one transistor or more. Hint: this may be related to your answer from question d).

[6]

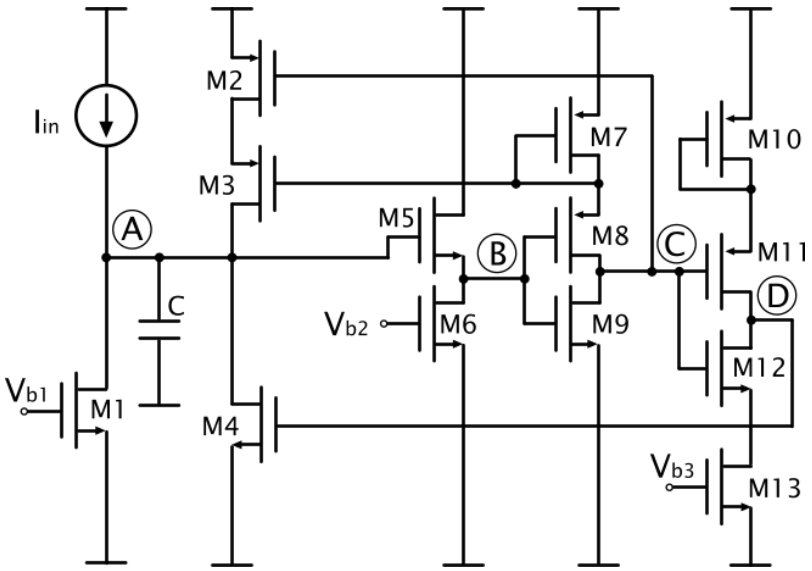


Figure 6.1

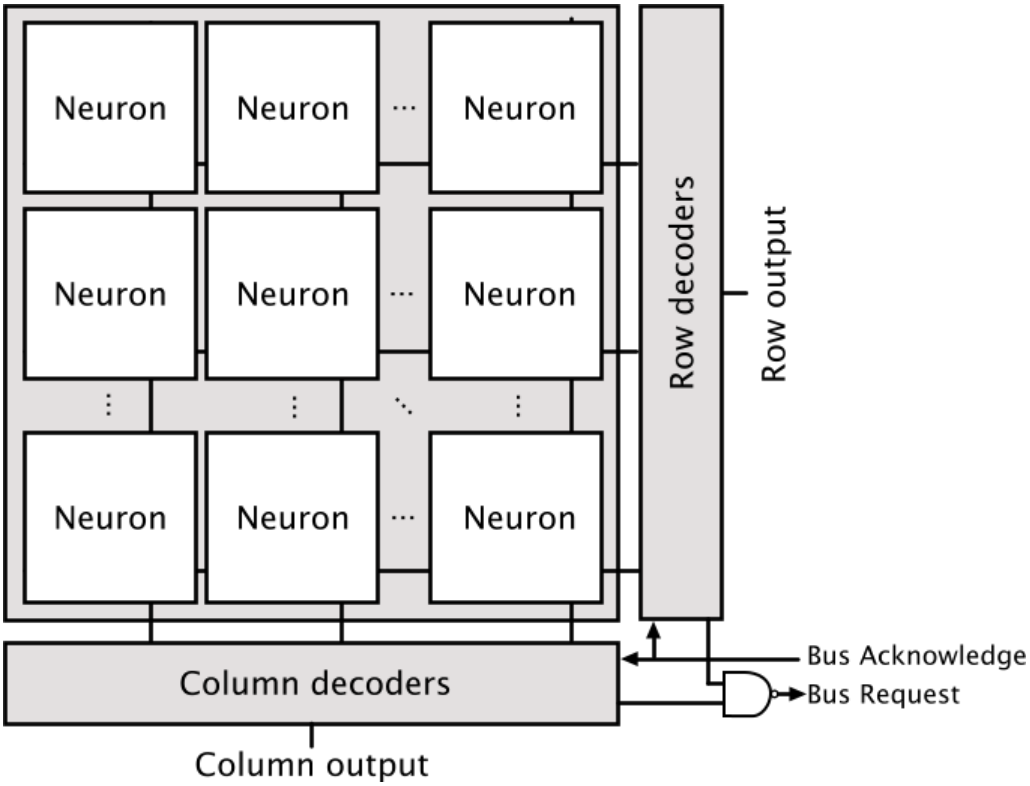

Neuron
Neuron
Neuron
Neuron
Neuron
Neuron
Neuron
Neuron
Neuron
Row decoders
Row output
Bus Acknowledge
Bus Request
Column decoders
Column output

Figure 6.2